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A Case Study of Functional Recovery for New Buildings

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https://doi.org/10.6028/NIST.TN.2351

NIST NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY
U.S. DEPARTMENT OF COMMERCE

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A Case Study of Functional Recovery for New Buildings

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This publication is available free of charge from:
https://doi.org/10.6028/NIST.TN.2351

September 2025



U.S. Department of Commerce
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NIST TN 2351
September 2025

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Copyright, Use, and Licensing Statements
NIST Technical Series Publication Identifier Syntax

Publication History
Approved by the NIST Editorial Review Board on 2025-09-11

How to Cite this NIST Technical Series Publication
Elsibaie, S., et al. (2025) Estimating the Costs and Performance of Enhanced Building Design: A Case Study of Functional Recovery for New Buildings. (National Institute of Standards and Technology, Gaithersburg, MD), NIST Technical Note (TN) NIST TN 2351. https://doi.org/10.6028/NIST.TN.2351

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Abstract

This report details the development of different structural design building archetypes and other seismic enhancements, the cost estimation process applied to those archetypes, and the economic implications and estimation of differing functional recovery times of the structural design choices after earthquake events. This culminates in a comparative benefit-cost analysis (BCA) of damage and recovery after earthquake events across the different structural archetypes and varying levels of structural stiffness. Using a framework that integrates civil engineering principles with economic evaluation, the research quantifies the economic implications of differences in performance of common structural systems—such as steel and concrete moment-resisting frames, shear walls, and braced frames, at different structural and non-structural intervention levels—under seismic loads. The cost estimation development process is described, and comparisons are made between industry and commercial construction cost data sources. The report lays the groundwork for the quantification of the trade-offs between initial construction costs, potential earthquake-induced damages, and economic benefits. The results suggest substantial repair cost reduction from recovery-based design, although recovery time is not always improved. These insights provide actionable guidance for engineers and policymakers seeking to balance safety, resilience, and cost-efficiency in seismic design, ultimately contributing to more economically resilient infrastructure in earthquake-prone regions.

Keywords

building design, cost estimation, functional recovery, re-occupancy, seismic design

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Acknowledgments

We are grateful to HB Risk Group for providing technical support for the development of the case studies. We thank David Butry, Christina Gore, and M. Amin Hariri-Ardebili for their review and the NIST Engineering Laboratory for 2021 Exploratory Project funding that provided the foundation for this report. All errors are our own.

1. Introduction

This report presents an end-to-end process for estimating construction costs for new buildings using commercially available data. Through a set of case study archetype buildings, the development of both structural and nonstructural construction costs associated with various design options was explored. The goal of the study is to compare costs of designing buildings for enhanced post-earthquake performance, for example the cost of designing to achieve functional recovery performance objectives[1] that go beyond life safety[2]. Such cost estimates can be used for conducting economic evaluation of potential recovery-based design options for new construction as well as for exploration of potential design alternatives from a project perspective (either in the planning stage or the design stage).

A natural question is to what degree such cost estimates are uncertain.. In order to quantify variance for different estimate methods, the cost estimates obtained through the methodology are compared with those obtained from a professional cost estimator. Our analysis shows that commercially available data is a reasonable and inexpensive alternative for cost estimation, up to limitations on cost categories that cannot be estimated from such data.

The case studies are not intended to be exhaustive or suggestive of a design recommendation in practice, though the design options represent viable options to improve performance. The case study archetypes are used to illustrate the integration of engineering design and economic evaluation for functional recovery, as detailed in Fung et al. (2025) [1].

The objectives of this report are to:

1. Document archetype designs for reinforced concrete moment frame buildings, reinforced concrete shear wall buildings, steel buckling-restrained braced frame buildings, and steel moment frame buildings, as well as design interventions intended to improve post-earthquake performance;
2. Present an “end-to-end” (step by step) cost estimation approach based on “off the shelf” methods and data;
3. Benchmark construction costs obtained using commercial data with industry costs (developed and validated by a professional cost estimator);
4. Present comprehensive cost estimates for baseline and recovery-based designs that are consistent with industry practice; and
5. Present seismic assessment results of performance enhancements, including annual mean repair costs and recovery times.

While the report is focused on illustrating the cost estimation approach and the associated performance enhancements, in future work we conduct a complete benefit-cost analysis to

[1] Functional recovery is defined in Sec 1.1 Background on Functional Recovery.
[2] Life safety is also defined in Sec 1.1 Background on Functional Recovery.

quantify the tradeoff in direct and indirect benefits associated with enhanced performance and relative to the expected increase in upfront construction costs [1].

1.1. Background on Functional Recovery

The National Earthquake Hazards Reduction Program (NEHRP) suggested three stages of recovery (reoccupancy, functional recovery, and full recovery) based on the level of a building's functionality compared to its pre-earthquake performance [2,3]. Designing for functional recovery can improve life safety and community resilience [2,4], prevent buildings and structures from severe damage and lengthy repair [3,5], and reduce the likelihoods of displacement and business interruption in the aftermath of a seismic event [4,6,7].

The 2018 NEHRP reauthorization, Public Law (P.L.) 115-307, tasked the National Institute of Standards and Technology (NIST) and the Federal Emergency Management Agency (FEMA) with providing recommendations to Congress for improving the built environment and critical infrastructure to reflect performance goals in terms of post-earthquake reoccupancy and functional recovery times [8]. The NIST-FEMA (2021) Report defines reoccupancy and functional recovery as [3]:

> ***Reoccupancy*** *is a post-earthquake performance state in which a building is maintained, or restored, to allow safe re-entry for the purposes of providing shelter or protecting building contents.*
>
> ***Functional recovery*** *is a post-earthquake performance state in which a building is maintained, or restored, to safely and adequately support the basic intended functions associated with the pre-earthquake use or occupancy of the building.*

With increasing interest and demand for resilience from stakeholders, the NIST-FEMA special report articulated the need to define the recovery-based objectives for individual buildings in terms of acceptable recovery times, accounting for occupancy type or lifeline services [6,9]. The report also described the approaches for achieving functional recovery objectives for new and existing buildings, as well as lifeline systems. The methodology for assessing recovery time has been significantly enhanced in the last two decades [10,11]. The performance-based earthquake engineering (PBEE) framework provides a probabilistic approach that integrates a probabilistic hazard analysis, with a structural response assessment and a damage assessment of each structural and nonstructural component within the building for a given ground-shaking intensity [12,13].

The FEMA P-58 methodology advanced PBEE for general use by providing a comprehensive library of fragility curves and consequence functions for a wide range of structural and nonstructural components [14]. The resilience-based engineering design initiative (REDi) rating system further improved FEMA P-58 by assigning repair classes to building components, incorporating impeding factors into repair time estimation, and defining three levels of functionality (reoccupancy, functional recovery, and full functionality) [15].

Both the framework proposed by Cook et al. [19] and the F-Rec framework [16, 17, 18] refined the modeling of re-occupancy and functional recovery times for modern U.S. buildings by employing fault trees to explicitly map component-level damage to system-level operations and then to building-level performance states. The Applied Technology Center (ATC) 138 project [20] adopted the functional recovery assessment framework proposed by Cook et al. [19], providing a combination of methods for assessing recovery time (ATC 2021). The ATC 138 methodology is still under development, with an objective to inform the development of future functional recovery provisions [20].

The literature to improve building recovery performance has increased dramatically in recent years, and the scope of research has been expanded from the building itself to the building and its connected utility or lifeline systems. For new buildings, design strategies for structural and nonstructural systems have been examined by several studies, ranging from concrete structures [5,21] to steel structures [22,23]. Notably, the American Concrete Institute (ACI) 374A committee is working to develop prescriptive requirements for concrete buildings designed for functional recovery [24]. The prescriptive requirements may be included in the 2026 NEHRP seismic provisions.

For existing buildings, studies have examined the feasibility for concrete buildings to attain functional recovery goals through seismic retrofits [25]. Overall, great challenges exist for existing buildings to achieve performance objectives above life safety. Life safety refers to the set of design principles and construction practices aimed at protecting the building's occupants from serious injury or death during and immediately after an earthquake. This involves ensuring that the building's structure remains stable enough to prevent collapse, allowing people enough time to safely evacuate. It includes the use of materials and construction techniques that can absorb and withstand seismic forces, as well as incorporating safety features like emergency exits and clear evacuation routes. The goal is not necessarily to avoid all damage to the building, but to preserve life by maintaining structural integrity long enough for safe evacuation.

The dependency between building functions and external infrastructure systems has been evaluated by a few studies. Mohammadgholibeyki et al. [26] suggested that for office buildings, the impact of disruption of utility systems depends on tenant’s requirements, which could be small for reoccupancy but significant if full service is required.

Furthermore, economic and social feasibility analyses have been performed to support the development of design standards for functional recovery. Abrahams et al. [9] identified cost, benefit, feasibility, effectiveness, and equity as the most relevant criteria for stakeholders when assessing and comparing design options for post-earthquake functional recovery. Fung et al. (2022) [4] developed an economic evaluation framework that includes methods for quantifying the potential benefits and costs of recovery-based building design. In a subsequent study, Fung et al. [27] discussed the economic mechanism for policy makers, decision makers, and end users to support the formulation, implementation, and evaluation of recovery-based design standards. Moreover, Echeverria et al. [28] outlined the barriers and opportunities for achieving functional recovery through seismic retrofit of existing buildings. According to this study, the

main barriers identified by stakeholders include retrofit costs, insufficient or inadequate tools and guidelines for retrofit implementation, difficulty in communication of benefits, and contradiction between community-scale goals and building-scale decisions.

1.2. Scope of this report

The purpose of this report is to: document the archetype baseline designs as well as the recovery-based design interventions; benchmark commercially available cost-estimation databases with industry costs; and present an overview of cost estimates and seismic performance for enhanced, recovery-based design options relative to code-conforming baseline designs.

More thorough economic evaluation to compare benefits and costs of enhanced design is beyond the scope of this report. We refer to the reader to the forthcoming study (Fung et al. 2025) [1] that illustrates the integration of economic evaluation into design, based on the framework in Fung et al. (2022) [4].

Moreover, the report illustrates asset-level design interventions, their costs, and performance, for a single site (Los Angeles) and a single occupancy class (commercial office). As such, the report is a case study that should not be interpreted as providing cost or performance results for other assets (e.g., in other sites) or occupancy classes.

The report is structured as follows. Section 2 describes the building archetypes. In Section 3, we describe cost estimation based on commercially-available cost data, provide comparisons to industry costs, and present complete cost estimates for each archetype. Section 4 describes the performance evaluation and provides seismic performance results. In Section 5, we discuss limitations and future work.

2. Building Archetypes

This section overviews the design details of a set archetype building models developed for this study. The archetypes were developed so that each was using one of four primary lateral force resisting systems (LRFS): Reinforced Concrete Moment Frame (RCMF), Reinforced Concrete Shear Wall (RCSW), Buckling-Restrained Braced Frame (BRBF), or Steel Moment Frame (SMF).

The archetypes are designed to meet the requirements prescribed in American Society of Civil Engineers (ASCE) 7-16 and applicable material standards and generally follow typical design practices in the U.S. We designed the archetypes for a hypothetical ASCE 7 Seismic Design Category D_{max} condition with a short period spectral acceleration of S_{DS} = 1.0 g and one-second period spectral acceleration of S_{D1} = 0.6g, as defined in FEMA P-695 [32]; the site's seismicity is assumed to be representative of the high seismicity, coastal California region, with soil conditions classified as very dense sand or hard clay (Site Class C of ASCE/Structural Engineering Institute (SEI) 7-16 Table 20.3-1). The hazard at the site is characterized by several faults and is not expected to exhibit near-field directivity effects.

The archetypes range in height from 4 to 20 stories, all with a rectangular floor plan, evenly distributed lateral load paths (i.e., limited torsion), and no vertical or horizontal irregularities. All archetypes are designed as commercial offices with typical floor live loads of 50 lb/sqft (2.4 kPa) and roof live loads of 20 lb/sqft (0.96 kPa). The layout of the various nonstructural components and systems with the building are allocated based on typical U.S. commercial office configurations as suggested in FEMA P-58-3 [31] and discussed in more detail in Section 3.4.

It should be noted that the 20-story RCSW and 18-story BRBF designs exceed the 160 ft height limit for Seismic Design Category D sites prescribed in Table 12.2-1 of ASCE 7-16. Therefore, to meet code requirements, these designs would need to be peer-reviewed to demonstrate compliance with the collapse performance objectives identified in the building code. However, nonlinear design and peer review are considered outside the scope of this study.

To study the change in construction cost and seismic response of buildings that are designed to meet enhanced post-earthquake performance with respect to reoccupancy and functional recovery times, each baseline (i.e., ASCE/SEI 7-16 compliant) archetype is redesigned considering one of three recovery-based design strategies (i.e., "design interventions"), namely: Nonstructural (nonstructural improvement only), Structural (structural improvement only), and Full (combined structural and nonstructural improvement). For the Nonstructural Improvement design intervention, a suite of nonstructural design alterations and enhancements is applied to several architectural and mechanical, electrical, and plumbing (MEP) systems within the building, and includes testing of nonstructural equipment, replacing vulnerable systems, and detailing connections for higher level of strength or deformation capacity. Specific details on the nonstructural improvements considered are discussed in Section 3.5.3.

For most of the structural system, the Structural Improvement design intervention involves increasing the required minimum base shear strength by 50 % (I_e = 1.5) and reducing the allowable peak story drifts from 2 % to 1 %, generally corresponding to the Risk Category IV

structural requirements of ASCE 7-16. However, the Structural Improvement design intervention of the BFBF system is achieved by coupling the BRBF with a moment-resisting frame (MRF) to form a dual system. The MRF, acting as a backup frame, provides the system with additional lateral stiffness and prevents large story drifts after BRBs yield [40]. Beyond the addition of the backup moment frame, no additional stiffness or strength requirements are considered.

The following sections present a detailed discussion of the baseline design and recovery-based design variants of each structural system. Additional details, such as member sizing and reinforcement layouts, for all archetypes considered can be found in the associated GitHub repository [41].

2.1. Reinforced Concrete Moment Frame Buildings

Reinforced concrete moment frames (RCMF) are lateral force resisting structural systems (LRFS) comprised of reinforced concrete beams, columns, and beam-column joints, and can be apportioned and sized to help structures resist earthquake loads [30]. Following special ductile detailing requirements typical in high seismic areas, RCMFs are typically designed to dissipate earthquake energy through nonlinear yielding at beam ends, relying on principles of strong-column-weak-beam, ductile detailing, and the redistribution of lateral force to other elements [30].

This study considers a set of four baseline special RCMF archetypal commercial office buildings of various heights, representing modern U.S. design and construction practices for high seismic regions. The buildings have a 120 ft x 120 ft (approx. 11.15 m^2) footprint with a 20 ft (6.1 m) bay spacing (six bays per side) and a 15 ft (4.6 m) first story height with 13 ft (4.0 m) typical above that. The LFRS is arranged in a perimeter frame configuration, with moment-resisting frames extending the complete length of each side, as shown in Figure 1. The gravity system consists of an 8-inch (0.2 m) thick reinforced concrete slab supported by 25 gravity columns on each story. Concrete compressive strengths are specified as 7 ksi (≈ 48,000 kPa) for columns and joints and 5 ksi (≈ 34,000 kPa) for all other elements.

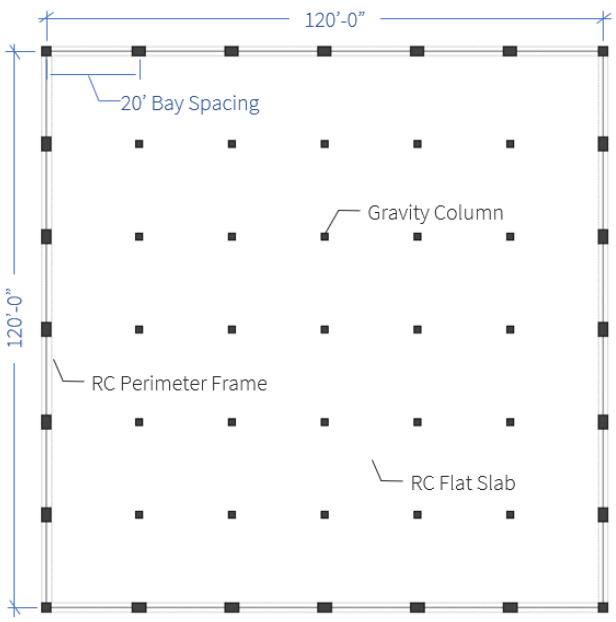


Figure 1. Typical plan view of the RCMF archetype. The LFRS occupies all bays on all sides of the building.

Earthquake loads are distributed to the RCMF system according to the equivalent lateral force procedure (ELFP) outlined in Chapter 12 of ASCE 7-16, with a response modification coefficient (R) of 8 and a deflection amplification factor (C_d) of 5.5, as specified in Table 12.2-6 of ASCE 7-16.

The four baseline buildings are redesigned considering the three recovery-based design strategies discussed above, resulting in 16 total RCMF archetypes. Table 1 outlines key design attributes for each RCMF baseline and variant archetype considered in this study.

Table 1. Design attributes of the RCMF archetype variants. (1 ft ≈ 0.3 m, 1 sqft ≈ 0.093 m^2)

Number of Stories	Design	Building Height (ft)	Usable Floor Area (sqft)	ASCE 7 Risk Category	Importance Factor, I_e	Allowable Design Drift	Building Period, T (s)
4	Baseline	54	57 600	II	1	2 %	0.98
4	Nonstructural	54	57 600	II	1	2 %	0.98
4	Structural	54	57 600	IV	1.5	1 %	0.67
4	Full	54	57 600	IV	1.5	1 %	0.67
8	Baseline	106	115 200	II	1	2 %	1.6
8	Nonstructural	106	115 200	II	1	2 %	1.6
8	Structural	106	115 200	IV	1.5	1 %	1.27

8	Full	106	115 200	IV	1.5	1 %	1.27
12	Baseline	158	172 800	II	1	2 %	2.23
12	Nonstructural	158	172 800	II	1	2 %	2.23
12	Structural	158	172 800	IV	1.5	1 %	1.62
12	Full	158	172 800	IV	1.5	1 %	1.62
20	Baseline	288	288 000	II	1	2 %	2.9
20	Nonstructural	288	288 000	II	1	2 %	2.9
20	Structural	288	288 000	IV	1.5	1 %	2.02
20	Full	288	288 000	IV	1.5	1 %	2.02

2.2. Reinforced Concrete Shear Wall Buildings

Reinforced concrete shear walls (RCSWs) are perhaps the most common concrete-based LRFS used in the U.S. and around the world, especially for areas with high seismic or wind loading conditions [37]. When detailed appropriately, shear wall elements can exhibit ductile failure modes desirable for dissipating energy from ground shaking.

This study considers a set of four baseline special RCSW archetypal commercial office buildings of various heights, representing modern U.S. design and construction practices for high seismic regions. The buildings have a 120 ft x 120 ft (11.1 m^2) footprint with a 30 ft (9.1 m) bay spacing (four bays per side) and a 15 ft (4.6 m) first story height with 13 ft (4.0 m) typical above that. The RC walls in the LFRS are arranged equally around the perimeter of the building, with one wall on each side, spanning the entire height of the structure, as shown in Figure 2. The shear walls are continuous up the height of the building without significant punchouts, but reduce in length with reduced loading conditions in the upper stories. The gravity system consists of a 12-inch (0.3 m) thick reinforced concrete slab supported by 25 gravity columns on each story. Concrete compressive strengths are specified as 6 ksi (≈ 41000 kPa) for the shear wall elements.

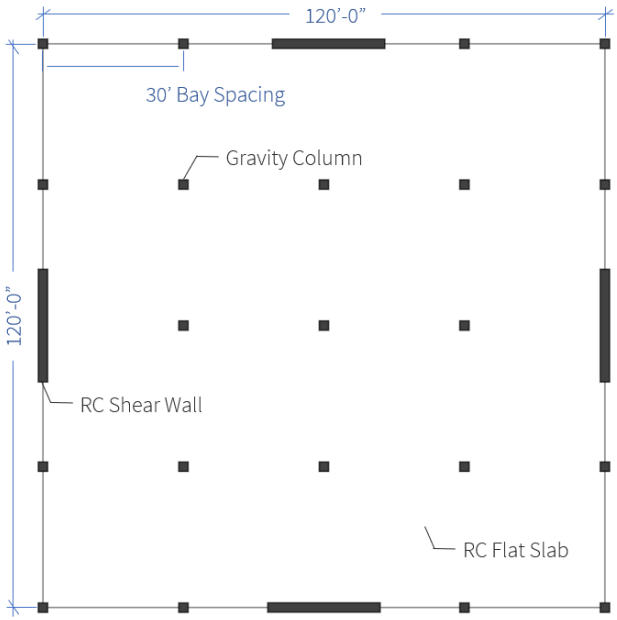


Figure 2. Typical plan view of the RCSW archetype. The shear walls on each side of the building span the entire height of the building, without discontinuity (though step down in length at the upper levels). (1' = 1 ft ≈ 0.3 m)

Earthquake loads are distributed to the RCMF system according to the modal response spectrum analysis (MRSA) outlined in Chapter 12 of ASCE 7-16, with a response modification coefficient (R) of 6 and a deflection amplification factor (C_d) of 5, as specified in Table 12.2-6 of ASCE 7-16.

The four baseline RCSW archetypes are redesigned considering the three recovery-based design strategies discussed above, resulting in 16 total RCSW design variants. Table 2 outlines key design attributes for each RCSW baseline and variant archetype considered in this study.

Table 2. Design attributes of the RCSW archetype variants. (1 ft ≈ 0.3 m, 1 sqft ≈ 0.093 m^2)

Number of Stories	Design	Building Height (ft)	Usable Floor Area (sqft)	ASCE 7 Risk Category	Importance Factor, I_e	Allowable Design Drift	Building Period, T (s)
4	Baseline	54	57 600	II	1	2 %	0.7
4	Nonstructural	54	57 600	II	1	2 %	0.7
4	Structural	54	57 600	IV	1.5	1 %	0.38
4	Full	54	57 600	IV	1.5	1 %	0.38
8	Baseline	106	115 200	II	1	2 %	1.62
8	Nonstructural	106	115 200	II	1	2 %	1.62
8	Structural	106	115 200	IV	1.5	1 %	0.45

8	Full	106	115 200	IV	1.5	1 %	0.45
12	Baseline	158	172 800	II	1	2 %	2.91
12	Nonstructural	158	172 800	II	1	2 %	2.91
12	Structural	158	172 800	IV	1.5	1 %	0.86
12	Full	158	172 800	IV	1.5	1 %	0.86
20	Baseline	288	288 000	II	1	2 %	3.9
20	Nonstructural	288	288 000	II	1	2 %	3.9
20	Structural	288	288 000	IV	1.5	1 %	1.7
20	Full	288	288 000	IV	1.5	1 %	1.7

2.3. Steel Buckling-Restrained Braced Frame Buildings

First introduced in the ASCE 7 design standard in 2005, buckling-restrained braced frames (BRBFs) are now used extensively in highly seismic regions [39]. In contrast to typical steel braces, which can buckle suddenly due to compressive loads, a buckling-restrained brace yields uniformly in compression and tension without buckling failure, allowing buildings to maintain stability at large displacement demands from earthquakes [39].

This study considers a set of three baseline BRBF archetypal commercial office buildings of various heights, representing modern U.S. design and construction practices for high seismic regions. The buildings have a 120 ft x 120 ft (11.1 m^2) footprint with a 30 ft (9.1 m) bay spacing (four bays per side) and a 13 ft (4.0 m) typical story height. The lateral braced frames are arranged equally around the perimeter of the building; the 4-story designs have a single 15 ft (4.6 m) bay, “lightning bolt” braced frame on each side of the building, whereas the 12 and 18 story designs have a double 15 ft bay “lightning bolt” braced frame configuration, as shown in Figure 3. The gravity system consists of a composite floor slab system consisting of a 3.5 inch (0.09 m) steel deck with a 2 inch (0.05 m) concrete topping slab supported by purlins and girders distributing vertical loads into 25 wide flange gravity columns on each story.

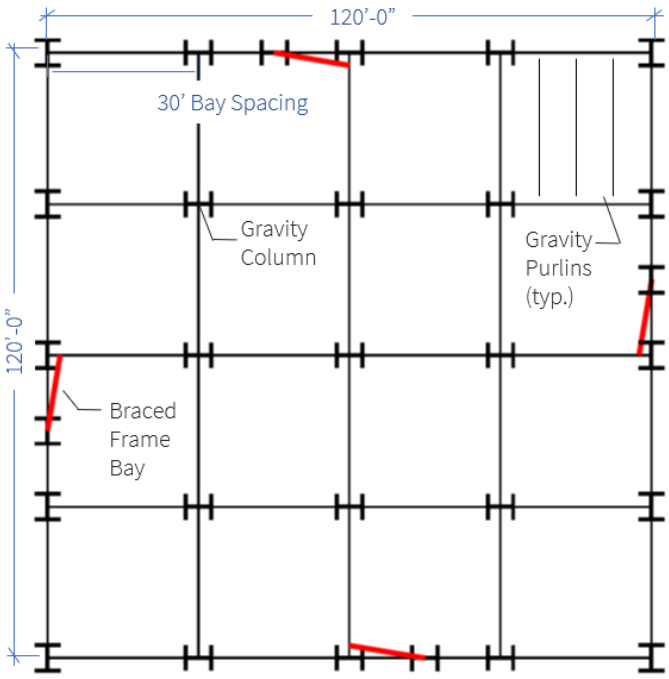


Figure 3. Typical plan view of the BRBF archetypes. For the four-story designs, the LFRS occupies a single 15' bay on each side of the building. For the 12 and 18 story design the LFRS occupies two 15' bays on each side. (1' = 1 ft ≈ 0.3m)

Earthquake loads are distributed to the BRBF system according to the modal response spectrum analysis (MRSA) outlined in Chapter 12 of ASCE 7-16, with a response modification coefficient (R) of 8 and a deflection amplification factor (C_d) of 5, as specified in Table 12.2-6 of ASCE 7-16.

The three baseline BRBF archetypes are redesigned considering the three recovery-based design strategies mentioned above, resulting in 12 total BRBF design variants. Table 3 outlines key design attributes for each BRBF baseline and variant archetype considered in this study.

Table 3. Design attributes of the BRBF archetype variants. (1 ft ≈ 0.3 m, 1 sqft ≈ 0.093 m^2)

Number of Stories	Design	Building Height (ft)	Usable Floor Area (sqft)	ASCE 7 Risk Category	Importance Factor, I_e	Allowable Design Drift	Building Period, T (s)
4	Baseline	54	57 600	II	1	2 %	0.70
4	Nonstructural	54	57 600	II	1	2 %	0.70
4	Structural	54	57 600	II	1	2 %	0.70
4	Full	54	57 600	II	1	2 %	0.70
12	Baseline	158	172 800	II	1	2 %	1.95

12	Nonstructural	158	172 800	II	1	2 %	1.95
12	Structural	158	172 800	II	1	2 %	1.95
12	Full	158	172 800	II	1	2 %	1.95
18	Baseline	288	259 200	II	1	2 %	2.64
18	Nonstructural	288	259 200	II	1	2 %	2.64
18	Structural	288	259 200	II	1	2 %	2.64
18	Full	288	259 200	II	1	2 %	2.64

2.4. Steel Moment Frame Buildings

Steel moment resisting frames (SMF) are another commonly used steel membered design element used in structures to resist lateral loading. Similar to RCMFs, SMFs are typically designed to dissipate earthquake energy through nonlinear yielding at beam ends. SMF structures have been used and studied intensely in earthquake prone areas such as Japan, United States and other regions [42,43].

This study considers a set of four baseline special SMF archetypal commercial office buildings of various heights, representing modern U.S. design and construction practices for high seismic regions. The buildings have a 120 ft x 120 ft (11.1 m^2) footprint with a 30 ft (9.1 m) bay spacing (four bays per side) and a 15 ft (4.6 m) first story height with 13 ft (4.0 m) typical above that. The designated LFRS frames are arranged equally around the perimeter of the building, typically spanning two to three bays per side, depending on the loading conditions, as shown in Figure 4. The gravity system consists of a composite floor slab system consisting of a 3.5 inch (0.09 m) steel deck with a 2 inch (0.05 m) concrete topping slab supported by purlins and girders distributing vertical loads into 25 wide flange gravity columns on each story.

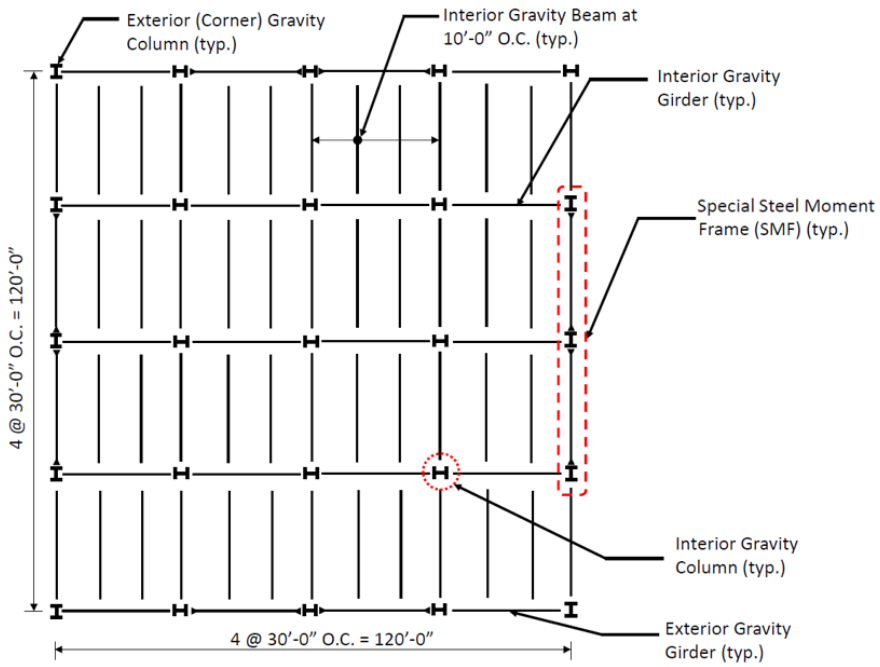


Figure 4. Typical plan view of the SMF archetypes. The LFRS occupies between two and three bays on each side of the building, depending on the building's size and design requirements. (1' = 1 ft ≈ 0.3 m)

Earthquake loads are distributed to the SMF system according to the modal response spectrum analysis (MRSA) outlined in Chapter 12 of ASCE 7-16, with a response modification coefficient (R) of 8 and a deflection amplification factor (C_d) of 5.5, as specified in Table 12.2-6 of ASCE 7-16.

The four baseline SMF archetypes are redesigned considering the three recovery-based design strategies discussed above, resulting in 16 total SMF design variants. Table 4 outlines key design attributes for each SMF baseline and variant archetype considered in this study.

Table 4. Design attributes of the SMF archetype variants. (1 ft ≈ 0.3 m, 1 sqft ≈ 0.093 m^2)

Number of Stories	Design	Building Height (ft)	Usable Floor Area (sqft)	ASCE 7 Risk Category	Importance Factor, I_e	Allowable Design Drift	Building Period, T (s)
4	Baseline	54	57 600	II	1	2 %	1.83
4	Nonstructural	54	57 600	II	1	2 %	1.83
4	Structural	54	57 600	IV	1.5	1 %	0.99
4	Full	54	57 600	IV	1.5	1 %	0.99
8	Baseline	106	115 200	II	1	2 %	2.31

8	Nonstructural	106	115 200	II	1	2 %	2.31
8	Structural	106	115 200	IV	1.5	1 %	1.61
8	Full	106	115 200	IV	1.5	1 %	1.61
12	Baseline	158	172 800	II	1	2 %	2.81
12	Nonstructural	158	172 800	II	1	2 %	2.81
12	Structural	158	172 800	IV	1.5	1 %	2.01
12	Full	158	172 800	IV	1.5	1 %	2.01
20	Baseline	288	288 000	II	1	2 %	3.62
20	Nonstructural	288	288 000	II	1	2 %	3.62
20	Structural	288	288 000	IV	1.5	1 %	2.57
20	Full	288	288 000	IV	1.5	1 %	2.57

3. Construction Cost Estimation Methodology

This section walks the reader through each step of cost estimation using commercially available construction cost data. The process is based on a construction cost database from a commercial vendor. RSMeans was used though alternatives such as Craftsman Cost books, Autodesk ProEst, and others are comparable and may offer similar information. Our goal is not to compare these products specifically. Our use of a specific product is not intended as an endorsement of a particular vendor but rather a case study that relies on commercially available construction cost databases [44].

While construction cost databases include materials, labor, and equipment costs for a broad range of unit costs, assemblies, and full building designs, we find that the level of granularity needed for evaluating performance of nonstructural components is not readily available from such databases, with examples discussed below. It is a natural question whether such cost databases, on their own or supplemented with additional publicly available data sources, are reliable for estimating construction costs associated with enhanced building performance. Some information regarding costs is sometimes not contained in the database and must be estimated based on prior project experience or specific local conditions and industry practices. We discuss some of the issues in this report. Thus, we benchmark our cost estimation methodology with industry costs, based on bottom-up costs developed by a professional cost estimator.

In the next subsection, we describe the set of relevant analysis parameters, in particular the base year, construction assumptions, and level of estimation detail. This determines the scope of the cost estimation analysis. We then describe the cost estimation process and compare it with cost estimates obtained from industry, including comparisons of cost elements that are available from the cost database.

3.1. Analysis Parameters

Several structure factors, time factors, and site parameters affect construction costs and must be considered in cost estimation models for the final monetary costs to be accurate and meaningful for those projects with similar features. Table 5 details some of the assumptions for parameters chosen for the construction cost modeling we performed.

Table 5. Construction Assumptions for a Selection of Analysis Parameters **(1' = 1 ft ≈ 0.3 m)**

Parameter	Assumption
Year of Construction	2023
Duration of Construction	the first year
Building Useful Life	50 years
Building Type	Commercial Office
Building Footprint	120'x120'
Site Location	Los Angeles
Site Class	"C"
Site Access	Congested (4 % cost premium)
Site Remoteness	Urban (0 % cost premium)

These analysis parameters influence the final costs and must be considered either using commercial cost databases or data from industry. Commercial cost database and other construction costs books contain detailed cost data specific to material quality, location, and time of purchase, so they can integrate several of our analysis parameters seamlessly. Some parameters, however, were not meaningful to apply to commercial cost databases without additional context from construction industry experience or professional guidance. For example, "congested" site access is an assumption we made due to our construction site location being in busy downtown Los Angeles. Using data and guidance from industry sources we placed a 4 % cost premium on all construction costs because of this assumption. Commercial cost databases do not have easily accessible data to estimate this kind of general cost premium due to our site assumptions, so this is an example of a limitation of using commercial cost databases alone to compute accurate construction costs. However, site access could be implicitly included when a user selects a region for analysis, but this is not clearly stated or granularly adjustable for the end user.

3.2. Project Definition

The scope of our initial commercial cost database estimate was limited by our project definition such that it required more data, planning, and construction details to approach the full cost of a realistic construction project. This is not unusual in practice, and in fact the American Society for Testing and Materials (ASTM) Standard E2516-11 [45] has set out a classification system for estimates which separates them into 5 categories, partially reproduced in Table 6, based on several criteria, but primarily based on the degree of project definition. These categories, including the primary characteristic, have overlap which are also noted. Our initial estimates with commercial cost databases were likely Class 4 or Class 5, sharing elements with a concept study but with some limitations based on the prototype building nature, but also elements of

greater estimation detail. Our projects had structural design details defined, but not a specific site chosen or fine architectural details and construction activities planned. Our initial estimates were therefore only based on the structural design details and other information that were known, and assumptions were able to be made to fill in other details. Equipment costs were included, but we went beyond stochastic modeling and largely used semi detailed unit costs.

Table 6. Primary Classification Criteria, Accuracy, and other selected data for estimate classes from the ASTM Generic Cost Estimate Classification Matrix, With Definition Overlap Noted. Source: ASTM [45].

Estimate Class	Primary Classification Criteria: Degree of Definition (% Complete)	Overlap with Higher Class	Accuracy Range Ratio (Class Accuracy +- Error / Class 1 +- Error)	Typical Methods
Class 5	0 % to 2 %	-	4 to 20	Stochastic
Class 4	1 % to 15 %	1 %	3 to 12	Primarily Stochastic
Class 3	10 % to 40 %	5 %	2 to 6	Mixed
Class 2	30 % to 70 %	10 %	1 to 3	Primarily Deterministic
Class 1	70 % to 100 %	0 %	1	Deterministic

With access to more resources and industry expertise in cost estimation, more detailed and documented assumptions about those details could be made. These details still generally lacked the definition to reach Class 1 or Class 2 level construction costs and were likely more consistent with Class 3 accuracy. However, because these estimation processes were applied to prototype buildings with limits to project definition, there are limitations and additional uncertainties that may cause the estimates to be Class 4, or even Class 5 in expansion to a larger population. To improve these results more project definition would be needed, and the process might include the preparation and engagement with designers, suppliers, and contractors necessary and contingent on constructing a real building at a certain location.

Commercial cost databases maintain lists of associated construction materials, assemblies and costs that consider labor and placement, as well as material costs by location and time of construction. This is a very useful set of data for a construction cost engineer to create an initial estimate for a project based off plan documents and a “material takeoff” or “quantity takeoff,” which is the total quantity of materials as well as equipment and labor expected to be needed for construction [46]. It was our experience that usage of commercial cost data requires it to be supplemented with additional information from construction planning resources or cost estimation professionals, for example for contractor overhead costs, permit costs, design costs, and more, to approach more refined construction costs for these types of projects.

The objective for our cost models is to compute the most accurate and realistic construction cost for a set of building archetypes as an input into a benefit-cost analysis (BCA) for post-earthquake functional recovery design for new construction. We therefore present the

commercial cost databases estimate comparisons to show what amount of deviation it has with industry sources. Estimated construction costs are predictions for projects that have yet to occur, and therefore actual costs of a project due to unexpected construction challenges, delays, labor shortages, change orders, and other factors could deviate from estimates to some degree.

For the final cost estimate, the estimation process considered the full scope estimate from industry sources, which includes material, labor, and placement costs, as well as contractor fees, utility fees, overhead, profit, permitting, contingency fees, administration services, design and engineering, site preparation, insurance and bonding, and other project costs [1]. Thus, to compare commercial cost databases data to industry data, we will limit the scope of both to the areas with overlap: superstructure material, labor, and placement costs.

3.3. Non-Structural Components

To estimate construction costs as well as expected functional recovery times, each archetype building is populated with an inventory of building components representative of modern U.S. construction for commercial office buildings. Specific component populations vary in quantity for different building heights, have minor variations in actual components for different building archetypes of the same height, and are shown for a 4-story baseline RCMF design in Table 7. Default component populations are based on the structural layouts and design of the archetype building and data from industry. Component fragility and consequence functions used to simulate component-level damage and consequences come from the FEMA P-58 fragility database [31].

Table 7. Nonstructural Component details for the 4-story baseline building. (1 ft ≈ 0.3 m, 1 sqft ≈ 0.093 m^2)

Component class (ASTM E1557)	**Category**	**Subcategory**	**Quantity**	**Units**
B2010 Exterior Walls	Exterior Walls	Exterior wall framing	29 808	sqft
		Exterior finish, brick or similar	23 733	sqft
		Insulation	23 733	sqft
		Gypsum board to interior face	23 733	sqft
	Roof screen	Cantilevered roof screen at mechanical	2 240	sqft
B2010 Windows	Aluminum framed windows		6 075	sqft

B3010 Roof Coverings	Roofing & Insulation	Flat roof	14 400	sqft
C1010 Partitions	Partitions	Gypsum board on metal stud partition	54 432	sqft
		Gypsum board only on struct stud- 1 side	0	sqft
C1020 Interior Doors	Doors	Single interior doors	202	EA
C1030 Fittings	Built in cabinetry and fittings	Commercial Office	57 600	sqft
C2010 Stair Construction	Stairs	General stairs	7	EA
		Primary circulation stairs	1	EA
C2020 Stair Finishes		General stairs	7	EA
C3010 Wall Finishes	Walls	Paint to gypsum board/CMU	96 300	sqft
C3020 Floor Finishes	Flooring	Commercial Office	57 600	sqft
C3030 Ceiling Finishes	Ceilings	Commercial Office	57 600	sqft
D1010 Elevators & Lifts	Elevators	Low-rise	0	EA
		High-rise - base	3	EA
		Added stops	0	EA
D2010 Domestic Plumbing Fixtures	Fixtures	Domestic plumbing fixtures	46	EA
D2020 Domestic Water Distribution		Included with fixture	46	EA
D2030 Sanitary Waste		Included with fixture	46	EA
D2040 Rain Water Drainage	Rain water drainage	Roof drainage - flat	14 400	sqft
		Roof drainage - flat - extra for height	0	ft

		Roof drainage - sloped	0	sqft
D2090 Other Plumbing Systems		Special Systems	57 600	sqft
D3040 Distribution Systems	Air supply	Complete HVAC system	57 600	sqft
		Air Handlers	57 600	sqft
		Fans	57 600	sqft
		VAV boxes	57 600	sqft
		Small ductwork	57 600	sqft
D3060 Controls and Instrumentation	Controls	General building controls	57 600	sqft
D4010 Sprinklers	Automatic Sprinkler systems	Ordinary Hazard sprinkler system	57 600	sqft
D5010 Electrical Service & Distribution	Service Voltage Distribution	Primary switchgear and transformers	57 600	sqft
D5020 Lighting & Branch Wiring	Lighting, fixture and wiring	Commercial Office	57 600	sqft
	User Power	Commercial Office	57 600	sqft
	Equipment Connections	Commercial Office	57 600	sqft
D5030 Communications & Security	Communication wiring & conduit	Telephone/Data	57 600	sqft
E1020 Institutional Equipment	Built-in equipment (Group 1)	Commercial Office	57 600	sqft

Notes: EA = Each (Priced as complete assembly);

3.4. Cost Estimation: Commercially-Available Data

We now describe our cost estimation process using commercially-available construction cost data. The purpose is to provide users with a “how-to” guide to using construction cost databases, illustrated with commercial cost databases, together with other industry information to develop their own cost estimates.

3.4.1. Cost Data

In this study, structural costs are estimated using material cost data for year 2023 construction in Los Angeles, California. A selection of construction, site and location parameters assumptions are detailed in Table 8.

As previously discussed, commercial cost databases include the costs of components, materials, associated labor and placement costs, and equipment. The database also includes some costs for contractor overhead and profit (though determining exactly how to apply this data requires expertise).

Professional cost estimation methods were used to develop bottom-up industry cost estimates that take building design parameters and construction, site and location parameters as inputs, to produce a comprehensive cost estimate that includes materials, labor, equipment costs, contractor overhead and profit, costs for building design, permit application, plan evaluation, site inspection, and transportation of materials and equipment.

As we will show in the next subsection, the commercial cost database process does produce similar results to the industry costs, conditional on both having a similar scope. However, the more limited scope we initially followed using the commercial cost database generated estimate does not account for some items that should be included to produce a more complete cost estimate for a structure.

Table 8. Scope of estimates, and assumptions

Cost	**Commercial cost data**	**Industry costs**	**Cost Assumptions**
Materials	x	x	Commercial Office Quality Materials
Labor	x	x	Standard CA Labor Rates
Equipment	x	x	Fair Site Access for Equipment
Project Costs, Fees, Contractor overhead and profit		x	Typical Project Costs from Industry
Design, permits, transportation, etc.		x	Urban Location

Commercial cost data is not used for nonstructural cost estimates because it requires design details to determine the unit prices of building components. For instance, commercial cost database assemblies for precast concrete panels require information on thickness, finishes, panel size, and insulation. However, these features are unrelated to seismic performance. Diving into these assembly details would make it more difficult to generalize the results in this report and of the subsequent economic evaluation [1]. Due to this limitation, for nonstructural components of interest for enhanced performance, we use replacement costs from FEMA P-58 to proxy for the cost of a new component. In this case, "replacement cost" means "repair cost" when the component is severely damaged and must be fully replaced.

Building repair (and, consequently, component repair) involves relocation, reinstallation, temporary protection, removal, replacement, and demolition of damaged components. In this regard, repair costs are lower than replacement costs when building components are slightly damaged and replacement is not required. However, when building components are severely damaged, repair costs can exceed replacement costs due to various repair activities in addition to replacing damaged components. This, together with the fact that FEMA P-58 cost data is depreciating in quality (it was collected over many years and released in 2012) mean that this approach to estimating nonstructural costs is suboptimal.

In contrast, industry professionals have expert knowledge and access to data that allows selection of typical default elements to be identified and chosen with limited design details as in this study. Industry costs for nonstructural components capture the typical new construction placement costs of their installation. This study did not account for the discrepancies between FEMA P-58 replacement costs and industry costs for new installation, but we note that future studies may explore this further to assist in providing some degree of cost estimation accuracy improvement with "off the shelf" methods.

Finally, while commercial cost databases provide users with the ability to price out structural improvements, the assembly approach for non-structural components does not provide the same level of granularity. Thus, it is not feasible to use "off the shelf" methods for nonstructural design enhancements. As we will discuss below, we developed cost deltas for nonstructural improvements through conversations with manufacturers (all errors are own). Due to the seismic specialization and uncommon use of some improvements, commercial cost data does not provide sufficient information for cost estimation. Thus, we rely on our heuristics for nonstructural improvements throughout this study and the subsequent economic evaluation [1].

3.4.2. Structural Cost Estimation Method

Structural costs. The baseline (and enhanced) structural costs reflect (increased) material and labor costs for a building constructed with a code-conforming (enhanced) structural system. Structural costs are computed as unit prices of structural components multiplied by quantities:

$$Structural\ costs\ = \sum [(Unit\ price\ of\ a\ structural\ component)\ \times Quantity]$$

Figure 5 illustrates the procedure used to estimate structural costs using commercial cost data. Step 1 inventories the type and quantity of structural components in the building. Step 2 looks up the commercial cost assembly data for each type of structural component. If there is a matched assembly in the database with respect to materials, size, and maximum capacity, the total cost of this component type can be estimated by multiplying assembly cost and quantity. Otherwise step 3 is taken to assemble the costs for the component using the commercial cost unit cost data, or alternatively interpolate the costs from two similar assemblies. Once the unit cost is determined, the total cost is estimated by multiplying unit cost and quantity.

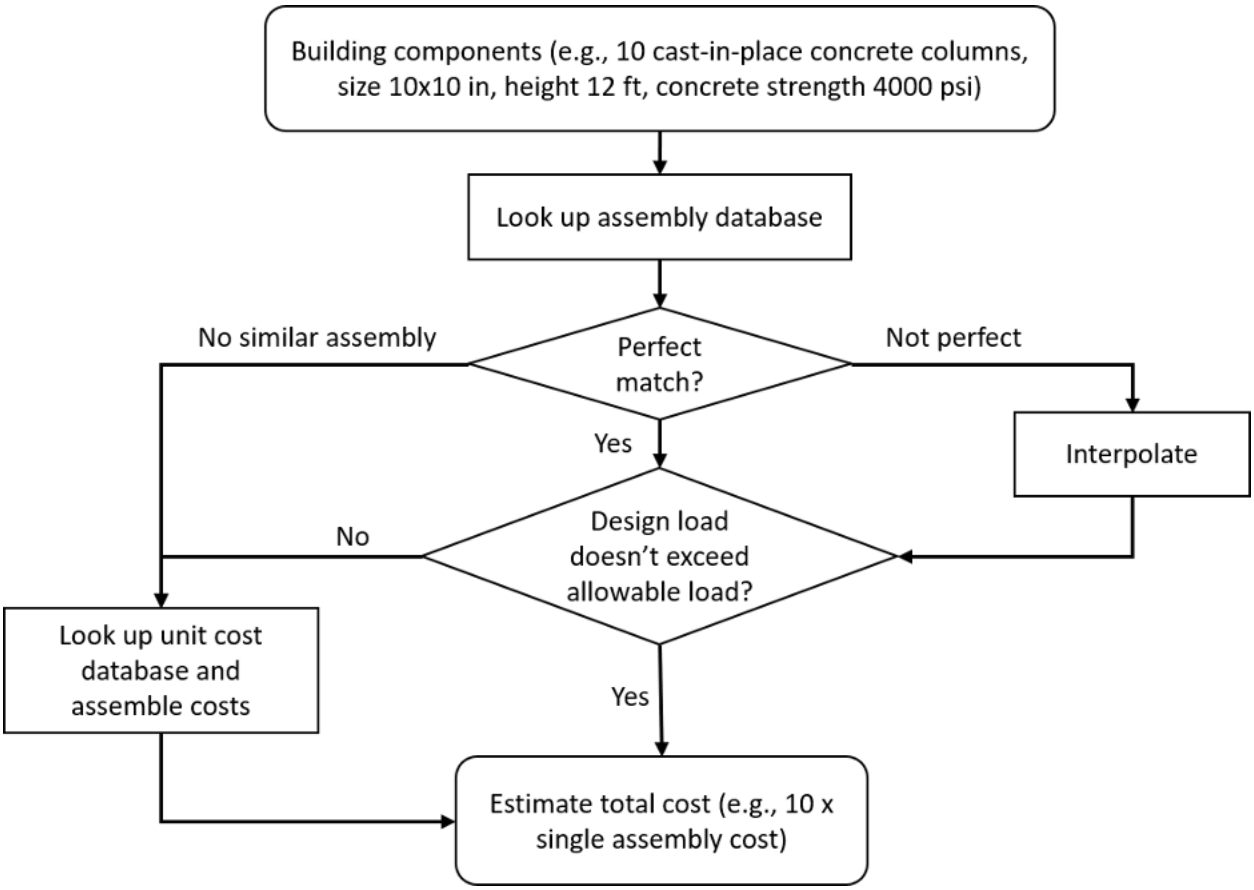


Figure 5. Commercial cost database component level estimation process. Source: Authors. **(12 in = 1 ft ≈ 0.3 m, 1psi ≈ 6895 Pa)**

As we will discuss in the Results section, this method produces similar costs to the professional cost estimator data within a comparable scope of structural elements and parameters. Note that the commercial cost databases database does not contain proprietary items used in building construction, such as the specific assemblies for buckling-restrained braces, which are fabricated by a small group of manufacturers [39]. The costs of proprietary items are estimated based on the information collected from fabricators, as will be discussed below.

As noted, BRBF components are not able to be priced generically in the same way the other primary construction materials can be. For BRBF structural elements, informed cost

breakdowns were utilized from industry data. The Connection cost is assumed to be 16 % of the total costs of columns, beams, and BRBs; the backup frame cost is assumed 27.5 % of the lateral system.

3.4.3. Nonstructural Cost Estimation Method

Nonstructural costs are computed as unit prices of nonstructural components times quantities:

$$Nonstructural\ costs\ = \sum [(Unit\ price\ of\ a\ nonstructural\ component) \times Quantity]$$

Nonstructural components of a typical commercial office building include stairs, elevators, cladding and glazing, ceilings, plumbing, fire protection, heating, ventilation, and air conditioning (HVAC) systems, and electrical systems.

Nonstructural improvement costs are estimated as the difference in nonstructural costs before and after upgrades. While a seismic assessment performance model will assume seismic performance characteristics (e.g., loads) for nonstructural components, it is not clear how that translates to actual industry components (and thus costs).

The Department of Health Care Access and Information (HCAI), formerly the Office of Statewide Health Planning and Development (OSHPD), leads a seismic certification program for nonstructural equipment (Special Seismic Certification Preapproval, OSP) and anchorages (Preapproval of Anchorage, OPA) used in health facilities in California. The preapproved equipment and anchorages can bear higher seismic loads and ensure continuing operation of hospitals after a major earthquake (HCAI 2022). Thus, we use OSP equipment and OPA anchorages in the recovery-based building design improvements.

The following cost assumptions are made with input from manufacturers and engineering consultants:

- Improvement cost for stairs is 13.5 % of stairs cost;
- Improvement cost for suspended ceilings is zero;
- Improvement cost for HVAC ducting, Variable Air Volume (VAV) box, HVAC fan, Air Handling Unit (AHU), electrical distribution panel, motor control center, low voltage switchgear, and transformer is 15 % of their original cost;
- Improvement cost for pipe bracing is 17.5 % of its original cost.

The resulting iterative and full (cumulative) nonstructural design improvements are shown in Table 9.

Table 9. Nonstructural configuration for recovery-based design, used across all structural systems**.**

#	System	Design Mitigation Action	Cost delta

1	Stairs	Add seismic joints to stair systems	13.5 %
2	Interior and fire suppression	Remove drop ceilings and heavy recessed lighting in favor of open concept ceilings and lightweight lighting. This also reduces damage to fire suppression systems from ceiling interactions.	0 %
3	HVAC system	Brace small HVAC ducting branches and drops which were previously unbraced	15 %
4	Plumbing system	Design pipe bracing to meet Ip = 1.5 requirements	17.5 %
5	HVAC system	Replace VAV box with OSP certified units	15 %
6	HVAC system	Replace exhaust fans with OSP certified units	15 %
7	HVAC system	Replace AHU with OSP certified unit and increase anchorage accordingly	15 %
8	Electrical system	Replace distribution panel with OSP certified unit and increase anchorage accordingly	15 %
9	Electrical system	Replace motor control center with OSP certified unit and increase anchorage accordingly	15 %
10	Electrical system	Replace low voltage switchgear with OSP certified unit and increase anchorage accordingly	15 %
11	Electrical system	Replace transformer with OSP certified unit and increase anchorage accordingly	15 %
12	Multiple systems	All of the above	Cumulative

Note: OSP = HCAI Special Seismic Certification Preapproval; HVAC = Heating, ventilation, and air conditioning; AHU = Air handling unit; Ip = Importance factor.

Generally, improving seismic performance for nonstructural components involves additional cost and certification, and this is reflected across most of the key non-structural components that are likely to be impacted by an earthquake. One exception is the suspended ceiling improvement, for which it is the actual removal of the typical drop ceiling component and replacement with an open ceiling design that facilitates the improvement in seismic performance.

In addition, we initially considered elevators for improved seismic performance but omitted them from the analysis as we were unable to obtain cost estimates for upgrades. Seismic design for elevators is complex, and the authors are unaware of any off-the-shelf enhancements that could be used for pricing. Determining the actual design enhancement and price data for elevators could be an enhancement in this methodology for future researchers. Additional building components that are not analyzed in this study include windows, doors, insulation, flooring, cabinets and countertops, lighting, and other appliances. These components are assumed to be identical for the baseline and enhanced buildings, and thus associated improvement costs are zero.

Moreover, maintenance costs excluding replacement, transportation costs, demolition, disposal, and recycling costs are assumed to be identical for all building designs. A replacement for nonstructural components in the middle of the analysis period is considered because the useful life of nonstructural components is between 15 and 30 years, less than that of structural components. This method is recommended by FEMA [31] when both structural and nonstructural components are analyzed in a BCA study. Accordingly, nonstructural improvement costs are counted twice in our cost estimation.

3.5. Benchmarking Construction Cost Estimates to Industry Costs for Superstructure Material Take-offs

With the scope limited so that a same-scope comparison may be made between the two construction cost data sources, it was found that the construction costs from commercial cost data tend to be relatively close to those from industry cost estimation. For a sample of our 4- and 12-story structures of each baseline (defined as conforming to ASCE 7-16) archetype, the cost differences of the commercial cost data estimate compared to the industry estimate detailed in Table 10 ranged from +28 % to -12 %. The cost samples and differences are summarized in the table below.

Table 10. Material, Labor, and Placement Cost Comparison of RSMeans Data and Professional Estimator

Archetype	Height	Commercial Cost Estimate	Industry Cost Estimate	% Difference
BRBF	4 story	$ 3 103 246	$ 2 704 000	15 %
	12 story	$ 11 924 881	$ 10 541 440	13 %
SMF	4 story	$ 3 444 890	$ 3 513 120	-2 %
	12 story	$ 12 149 168	$ 12 537 200	-3 %
RCMF	4 story	$ 4 040 941	$ 3 296 800	23 %
	12 story	$ 14 181 479	$ 11 117 600	28 %
RCSW	4 story	$ 3 056 176	$ 3 461 120	-12 %
	12 story	$ 10 750 801	$ 11 161 280	-4 %

This data shows the final costs for material take-off-based estimates from industry sources compared to commercial cost databases. The expected error for a Class 5 estimate is 4x to 20x greater than Class 1 error, while Class 3 estimates are 3x to 12x as shown in Table 6, but this information is not readily applicable to our comparison data shown above since neither a Class 1 estimate was prepared nor are the actual construction costs of the archetypes known. While it is feasible to reduce error through price gathering effort, the ultimate error for an estimate is knowable only after construction is in an advanced stage and all costs are tallied. . However, based on this sample of cost differences we believe commercial cost databases can be used to prepare similar estimation results compared to an industry estimate at least within the limited scope we have described. If the scope is desired to be increased, an industry professional familiar with local or regional construction practices should still be engaged to ensure all the additional miscellaneous construction and project costs previously described are considered to estimate the costs to fully construct the structures, as needed. In most cases the final total project costs based on industry costs were more than 10x the superstructure material take-off costs (see Table 11 Sec. 3.6).

3.6. Cost Estimation: Industry Costs

Finally, Table 11 presents estimated structural, nonstructural, total construction, and total project unit costs in dollars per square foot (1 square foot ≈ 0.093 m^2) for the baseline designs as well as the recovery-based design interventions, based on our industry cost estimates. The cost estimates shown in Table 11 cover a broader scope[3] than the cost estimates from commercially-available data as shown in Sec. 3.5[4] and thus provide a more complete picture of the relative unit costs across systems and stories, for recovery-based design relative to baseline designs.

At a glance, we can see the following trends for our case studies:

[3] As discussed in Sec. 3.2, between Class 3 and Class 4.
[4] As discussed in Sec. 3.2, between Class 4 and Class 5.

- Structural costs account for a smaller fraction (between 10 % and 15%) of the total cost than nonstructural costs (about 50 %)
- On average, recovery-based design increases structural costs by about 20 %
- On average, recovery-based design increases nonstructural costs by about 3 %
- Thus, because of the relative fractions of each in the total cost, each recovery-based design intervention increases total construction costs by between 2 % and 3 %, on average
- The full design intervention (structural plus nonstructural improvements) is additive and increases total construction cost by about 6 %, on average
- Given our assumptions, total project costs are directly proportional to total construction costs and do not account for potential increases in soft costs associated with enhanced performance

Table 11. Structural, nonstructural, total construction, and total project unit costs (in dollars per square foot) for each lateral force resisting system (System), number of stories (Stories), and design intervention (Design).

System	Stories	Design	C_S	C_{NS}	C_C	C_P
RCMF	4	baseline	72	351	527	696
RCMF	4	structural	87	351	545	719
RCMF	4	nonstructural	72	362	540	713
RCMF	4	full	87	362	557	736
RCMF	8	baseline	80	335	515	679
RCMF	8	structural	96	335	534	706
RCMF	8	nonstructural	80	345	527	696
RCMF	8	full	96	345	547	722
RCMF	12	baseline	78	329	505	667
RCMF	12	structural	92	329	523	691
RCMF	12	nonstructural	78	339	518	684
RCMF	12	full	92	339	536	707
RCMF	20	baseline	84	328	510	674
RCMF	20	structural	101	328	531	701
RCMF	20	nonstructural	84	338	523	690
RCMF	20	full	101	338	543	717
RCSW	4	baseline	75	351	531	701
RCSW	4	structural	86	351	544	719
RCSW	4	nonstructural	75	362	543	717
RCSW	4	full	86	362	557	735
RCSW	8	baseline	83	335	518	684
RCSW	8	structural	108	335	549	725
RCSW	8	nonstructural	83	345	531	701
RCSW	8	full	108	345	562	742
RCSW	12	baseline	78	329	506	668
RCSW	12	structural	98	329	530	700
RCSW	12	nonstructural	78	339	518	684
RCSW	12	full	98	339	543	716
RCSW	20	baseline	89	328	517	682
RCSW	20	structural	109	328	542	715
RCSW	20	nonstructural	89	338	529	699
RCSW	20	full	109	338	554	731
SMF	4	baseline	76	351	532	702
SMF	4	structural	92	351	551	728
SMF	4	nonstructural	76	362	544	719
SMF	4	full	92	362	564	744
SMF	8	baseline	86	335	522	689
SMF	8	structural	104	335	545	720
SMF	8	nonstructural	86	345	534	705
SMF	8	full	104	345	558	736
SMF	12	baseline	86	329	516	680
SMF	12	structural	107	329	542	715
SMF	12	nonstructural	86	339	528	697
SMF	12	full	107	339	554	732
SMF	20	baseline	105	328	537	709
SMF	20	structural	132	328	570	752
SMF	20	nonstructural	105	338	549	725

System	Stories	Design	C_S	C_{NS}	C_C	C_P
SMF	20	full	132	338	582	769
BRBF	4	baseline	62	349	512	675
BRBF	4	structural	70	349	521	688
BRBF	4	nonstructural	62	359	524	692
BRBF	4	full	70	359	534	705
BRBF	12	baseline	74	329	500	660
BRBF	12	structural	88	329	517	683
BRBF	12	nonstructural	74	339	513	677
BRBF	12	full	88	339	530	699
BRBF	18	baseline	81	328	507	670
BRBF	18	structural	96	328	526	695
BRBF	18	nonstructural	81	338	520	686
BRBF	18	full	96	338	539	711

C_S – Structural construction costs, including superstructure and foundation – Labor and Material (USD/SqFt)

C_{NS} – Nonstructural construction costs, not including owner furnished FF&E – Labor and Material (USD/SqFt)

C_C – Total construction costs, including site prep and contractor fees (USD/SqFt)

C_P – Total project costs – including soft costs such as designer fees, admin fees, and additional contingencies (USD/SqFt)

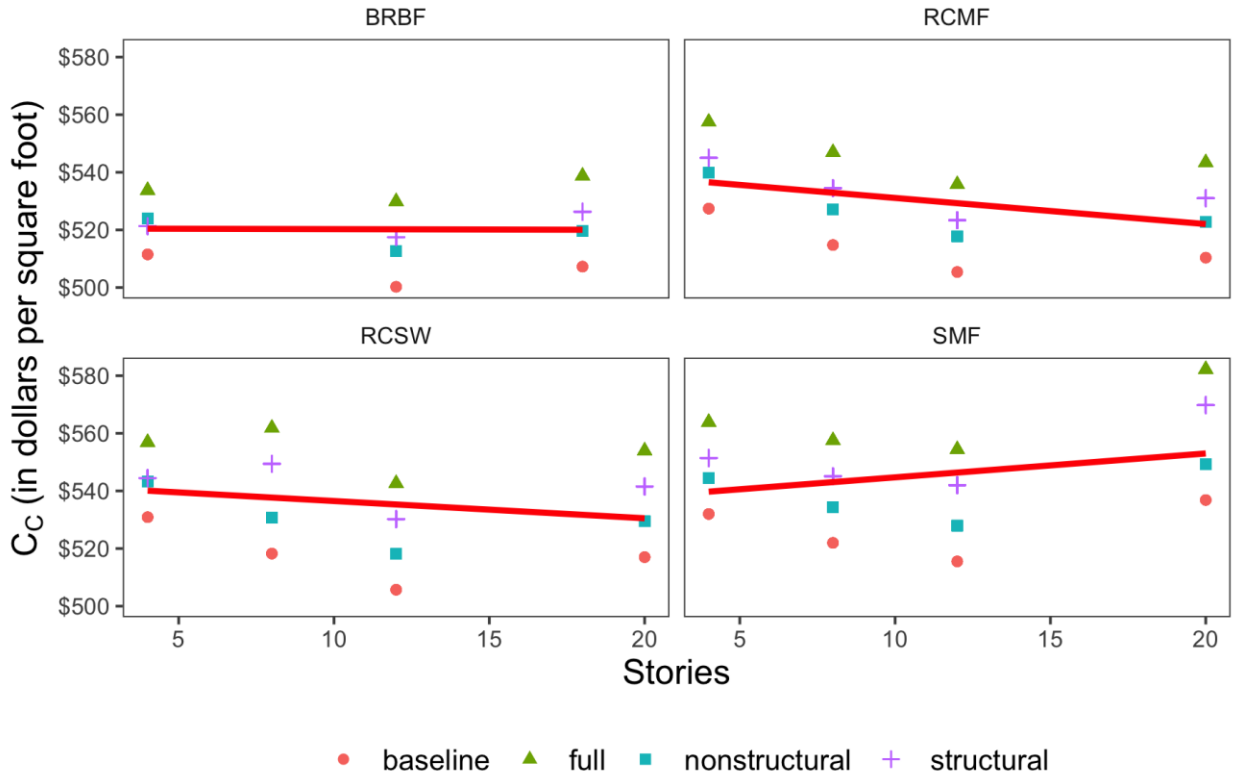


Figure 6. Total unit construction costs, C_C, for each design option, across each lateral force resisting system and across stories. The red line is a linear fit trend line.

Figure 6 plots the construction cost of each structural archetype, an output of the construction cost estimation process. The linear trend encompasses all data elements to identify trends in overall costs for each structural archetype separately and across all intervention types and building heights. Figure 7 plots the *cost deltas*, defined as the percentage increase in total construction cost over the baseline design, for each of the recovery-based design interventions, across lateral force resisting systems and stories. Similarly, the linear trend encompasses all data elements, to identify overall trends in costs deltas for each structural archetype separately and across all intervention types and building heights. The plots show cost deltas of 2 % and 4 %, which only increase moderately with height (and are pretty much flat for the concrete systems). Across all systems, the figure shows that cost deltas for nonstructural interventions do not increase with building height, and this makes intuitive sense because these members do not need to necessarily resist larger loads because of the increased building height. On the other hand, structural interventions do show increasing cost deltas with height for the steel systems (BRBF and SMF) and RCSW. The result is that cost deltas for the full intervention tends to increase with height. Finally, the figures illustrate the relationships between the cost deltas for each intervention. The cost deltas from structural interventions are consistently higher than cost deltas from nonstructural interventions, and the cost deltas for the full interventions

follow the trend for the structural interventions but are higher as the interventions combine structural and nonstructural improvements.

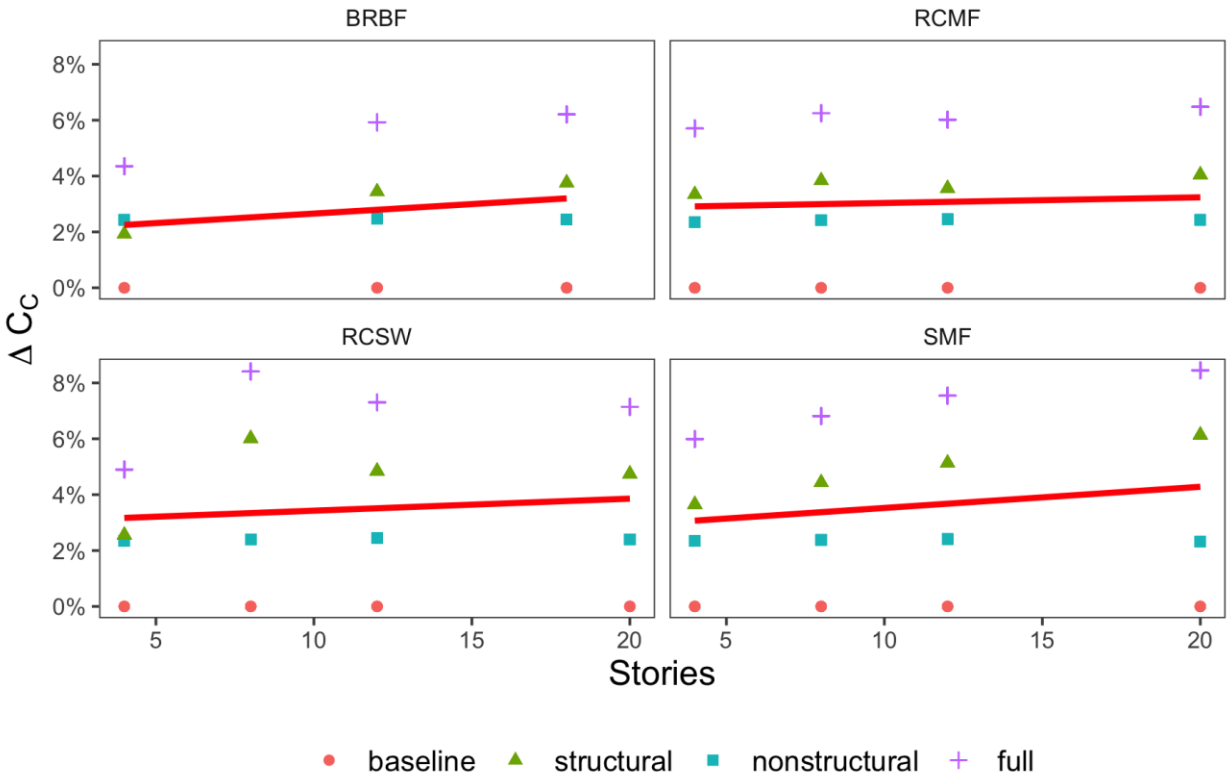


Figure 7. Percentage change in total construction cost, C_C, for structural, nonstructural, and full recovery-based. The red line is a linear fit trend line.

4. Functional Recovery Performance Evaluation

To evaluate the seismic performance of each archetype building in terms of repair cost and recovery time, we use the FEMA P-58 performance-based assessment methodology and the ATC 138 downtime estimation methodology [31].

4.1. Performance Based Methodology

Performance-based earthquake engineering (PBEE) is a statistically robust means of quantifying and communicating building performance into multiple metrics, such as repair costs or functional recovery. The approach integrates a probabilistic hazard analysis, with a structural response assessment and a component-level damage assessment to quantify the damage to each structural and nonstructural component within the building for a given shaking intensity (Porter 2003; Moehle and Deierlein 2004). The performance outcomes from a performance-based assessment depend on the building's specific design and configuration characteristics and can be used to facilitate the comparison of design alternatives and mitigation strategies for new recovery-based performance objectives.

The ATC 138 methodology is a widely used approach for quantifying building recovery performance. The framework integrates the functional recovery assessment framework proposed by Cook et al. (2022) into the FEMA P-58 methodology to establish a robust recovery-based performance evaluation approach alongside the other performance metrics considered in FEMA P-58, such as repair costs, repair time, casualties, and unsafe placards.

The analytical methodology is built on the performance-based earthquake engineering (PBEE) framework, which integrates a probabilistic hazard analysis, a structural response assessment, and a component-level damage assessment to quantify the damage to structural and nonstructural components of a building and aggregate them into expected building-level losses due to earthquakes (Porter 2003; Moehle and Deierlein 2004). In each step of the framework, results are simulated through a Monte Carlo process to account for uncertainties in ground shaking, building response, and component damage and repair.

In this study, we calculate building-level losses as Expected Annual Losses (EAL) to represent the seismic performance in terms of repair costs and recovery times. EAL are the anticipated losses, on average, one can expect to pay each year if losses due to earthquakes were distributed evenly per year.

$$EAL = \int_{edp} \int_{im} \sum_{DS} L_{DS} \times G(L|DS) dG(DS|edp)\, dG(edp|im) dP(im)$$

where EAL is expected annualized loss. L_{DS} is the repair cost for a component in a damage state DS. $G(L|DS)$ is a consequence function that links damage to loss (e.g., repair costs). $G(DS|edp)$ links damage to an engineering demand parameter (e.g., interstory drift ratio, peak ground acceleration). $G(edp|im)$ links an engineering demand parameter to a ground shaking

intensity. $P(im)$ is the exceedance probability, describing the annual probability that an earthquake exceeds an intensity level.

The following sections outline each step of the performance evaluation procedure used to estimate the expected annual repair costs and recovery times for each archetype building.

4.2. Site Hazards

We use expected spectral acceleration (Sa) values defined by the 2014 National Seismic Hazard Model (Petersen et al. 2014) to represent the seismic demands for each archetype across various levels of hazard intensity. While the design values used to design the archetypes are generic, the hazard is selected based on a specific site in Los Angeles, CA (Lat: 34.05; Long: -118.25) with similar seismicity. The site's seismicity is assumed to be representative of the high seismicity, coastal California region, with soil conditions classified as very dense sand or hard clay (Site Class C of ASCE/SEI 7-22 Table 20.2-1). The hazard at the site is characterized by several faults and is not expected to exhibit near-field directivity effects. The EALs presented here are specific to the relationship between the hazard at the selected site and the selected design values, and EALs will vary from site to site. Accordingly, the cost estimates for recovery-based designs can vary by building location as well.

4.3. Structural Response

To simulate the nonlinear response of the archetype buildings, we develop analytical models to represent the response characteristics of each baseline model and structural design variant. The models are developed in the open-source platform Opensees and represent the building response via 2-D, 3-bay representations (as there is no irregularity in the floor plan, half of the plan used for the analytical representation of archetypes) of the full building design similar to those used by Haselton et al. [33]. The strength and stiffness of the gravity system is represented within the properties of the lateral elements (e.g., added beam stiffness) but are not explicitly modeled.

Material nonlinearity of the structural elements in each model except RCSW is represented using a lumped-plasticity approach, while RCSW uses a distributed plasticity (MVLEM element) approach, each using ASCE 41 for modeling parameters. Elastic damping is represented by Rayleigh damping at 5 % of critical, applied to each models' first and third mode; hysteretic damping is accounted for in the nonlinear springs. Global P-Delta affects are accounted for via a leaning column. The nonlinear response results for the baseline models and structural variants can be found in the associated GitHub repository [41].

To predict the structural response of each archetype, we scaled a suite of 44 far-field ground motions, representative of large earthquakes in coastal California regions, to 14 ground shaking intensity levels, and collected the engineering demand parameters (EDP) associated with the component fragilities populated within each archetype's performance model, namely, peak

story drift (SDR), peak floor accelerations (PFA), peak wall chord rotations (CR), and peak residual story drifts (RSDR).

4.4. Component Damage and Building Loss

In the performance evaluation methodology, consequences are first determined by simulating damage to each structural and nonstructural component within the building, based on the particular response of the structure to the ground shaking.

To facilitate the component-level damage simulations, FEMA P-58 collects a database of over 700 component fragility and consequence functions for various structural and nonstructural building components based on empirical earthquake data, experimental testing, analytical analysis, and engineering judgment.

Each archetype building assessed in this study is populated with a suite of structural and nonstructural components from the FEMA P-58 database, containing component-specific fragility and consequence functions. Building level repair costs are the sum of the cost to repair each damaged component within the building.

To assess the functional recovery time of the building given component damage, the ATC 138 methodology uses a series of fault trees, which map the relationship between component-level damage to system-level operations and building occupancy and function. A repair scheduling algorithm is used to allocate workers to various repair needs throughout the building and determine when the building function will be restored. This also includes temporary repair scenarios in which repair actions only mitigate short-term impact of component and system failures. Additionally, a broader range of external factors that may delay the initiation of repairs are considered to reflect the current practice in the US, including inspection processes, permitting, securing financing, and mobilization of engineering teams and contractors.

4.5. Case Study Functional Recovery Performance Results

Table 12 provides the annualized seismic performance assessment results for the baseline design archetypes as well as the recovery-based design interventions, including both the mean, μ, and standard deviation, σ, repair cost and recovery time, $\boldsymbol{\tau(fr)}$, outcomes. Additionally, Table 13 provides the seismic performance assessment results for a ground motion shaking intensity equivalent to the ASCE 7 Seismic Design Category D_{max} condition (S_{DS} = 1.0 g and S_{D1} = 0.6g). The functional recovery time is the time measured in days for the repair and restoration activities that are necessary to restore function to the building. The distributions of repair cost and recovery time results are unlikely to follow a normal distribution. At a glance, we can see the following building performance trends for our case studies:

- Repair cost and recovery time outcomes are specific to a given structural system and building height.

- Structural improvements tend to reduce repair costs (28 % reduction in repair costs on average) but have a mixed impact on functional recovery time (ranges from a 31 % reduction to a 93 % increase in recovery time).
- Nonstructural improvements always reduce both repair cost (16 % mean reduction) and functional recovery time (30 % mean reduction).
- Combined structural and nonstructural improvements (full) always lead to the largest reduction in repair costs (54 % mean reduction), but not always the largest reduction in functional recovery time (27 % mean reduction).

In summary, functional recovery times are not always proportional to repair costs, and improvements made to the structural system and nonstructural systems have variable impacts on each archetype building. Structural interventions have the largest impact on repair costs while nonstructural interventions more substantially improve recovery time.

Damage to structural components results in expensive repairs but is less likely to cause disruptions overall, as severe structural damage only tends to occur at very large levels of shaking. Additionally, structural improvements tend to stiffen the building's structural response, resulting in increased accelerations. These accelerations damage nonstructural components, increasing the probability of loss of building function and offsetting benefits from reduced structural damage. An example of this is the 4-story RCSW building, where stiffening an already short and stiff structure provides little structural benefit while substantially increasing the forces on the nonstructural component within the building.

Damage to nonstructural components, on the other hand, tends to be less expensive but can still be quite disruptive. Compared to severe structural damage, nonstructural components are damaged more often at moderate levels of shaking. Additionally, nonstructural design improvements only enhance component capacity without any negative impacts on structural response.

Table 12. Numerical summary of expected performance for enhanced design (structural, nonstructural, and full interventions) relative to the baseline design, in terms of annualized repair cost (measured in dollars) and annualized functional recovery time ($\tau(fr)$, measured in days) for each structural system and story height.

System	Stories	Design	Repair Cost (USD)		$\tau(fr)$	
			μ	σ	μ	σ
RCMF	4	baseline	$39,012	$19,077	2.14	1.17
RCMF	4	nonstructural	$36,233	$18,041	1.52	0.90
RCMF	4	structural	$18,390	$11,055	2.04	1.33
RCMF	4	full	$13,756	$9,227	1.24	1.02
RCMF	8	baseline	$34,022	$21,582	2.05	1.26
RCMF	8	nonstructural	$30,749	$21,985	1.38	1.00
RCMF	8	structural	$31,052	$19,184	2.10	1.33
RCMF	8	full	$26,002	$17,978	1.31	1.01
RCMF	12	baseline	$30,993	$20,112	2.03	1.33
RCMF	12	nonstructural	$28,215	$19,269	1.42	1.08

System	Stories	Design	Repair Cost (USD)		$\tau(fr)$	
			μ	σ	μ	σ
RCMF	12	structural	$27,364	$17,706	2.20	1.48
RCMF	12	full	$22,718	$15,608	1.49	1.14
RCMF	20	baseline	$38,711	$26,546	2.26	1.36
RCMF	20	nonstructural	$31,906	$23,563	1.66	1.16
RCMF	20	structural	$28,854	$19,565	2.30	1.52
RCMF	20	full	$19,917	$13,487	1.58	1.20
RCSW	4	baseline	$10,833	$8,809	1.21	0.88
RCSW	4	nonstructural	$7,494	$6,969	0.75	0.63
RCSW	4	structural	$14,658	$11,213	2.03	1.30
RCSW	4	full	$6,899	$6,883	1.34	1.06
RCSW	8	baseline	$25,490	$16,421	1.50	0.84
RCSW	8	nonstructural	$21,704	$14,183	1.06	0.67
RCSW	8	structural	$26,346	$17,072	2.90	1.63
RCSW	8	full	$11,575	$7,374	2.07	1.31
RCSW	12	baseline	$39,208	$24,380	2.10	1.07
RCSW	12	nonstructural	$34,942	$21,898	1.50	0.80
RCSW	12	structural	$9,907	$7,821	1.66	1.29
RCSW	12	full	$4,594	$3,883	1.06	0.94
RCSW	20	baseline	$31,561	$20,833	1.64	0.86
RCSW	20	nonstructural	$26,875	$19,882	1.07	0.62
RCSW	20	structural	$19,428	$12,891	1.79	1.12
RCSW	20	full	$10,782	$8,994	1.10	0.88
SMF	4	baseline	$65,742	$33,743	2.20	0.86
SMF	4	nonstructural	$61,440	$31,690	1.64	0.69
SMF	4	structural	$16,042	$9,693	1.52	0.87
SMF	4	full	$10,429	$7,788	0.96	0.69
SMF	8	baseline	$35,645	$22,072	1.66	0.88
SMF	8	nonstructural	$31,776	$21,842	1.15	0.67
SMF	8	structural	$38,681	$22,989	2.24	1.24
SMF	8	full	$29,513	$20,115	1.52	0.92
SMF	12	baseline	$65,637	$37,554	1.75	0.83
SMF	12	nonstructural	$63,729	$39,866	1.28	0.64
SMF	12	structural	$28,989	$21,656	1.63	0.92
SMF	12	full	$23,189	$20,341	1.11	0.76
SMF	20	baseline	$44,658	$27,036	1.87	0.96
SMF	20	nonstructural	$35,777	$25,140	1.33	0.72
SMF	20	structural	$22,094	$14,605	1.40	0.89
SMF	20	full	$14,927	$12,134	1.04	0.76
BRBF	4	baseline	$31,582	$14,939	2.77	1.49
BRBF	4	nonstructural	$23,755	$14,034	1.86	1.18
BRBF	4	structural	$24,076	$11,805	2.75	1.50
BRBF	4	full	$14,680	$8,980	1.83	1.18
BRBF	12	baseline	$49,093	$27,729	2.92	1.52
BRBF	12	nonstructural	$42,614	$30,735	2.01	1.18
BRBF	12	structural	$32,405	$19,047	2.93	1.56
BRBF	12	full	$22,481	$16,931	2.02	1.23
BRBF	18	baseline	$38,798	$23,664	2.74	1.43
BRBF	18	nonstructural	$22,789	$18,474	2.00	1.26

System	Stories	Design	Repair Cost (USD)		$\boldsymbol{\tau(fr)}$	
			μ	σ	μ	σ
BRBF	18	structural	$33,449	$18,968	2.72	1.44
BRBF	18	full	$16,550	$9,427	1.98	1.25

Table 13. Numerical summary of expected performance for enhanced design (structural, nonstructural, and full interventions) relative to the baseline design, in terms of repair cost (measured in millions of dollars) and functional recovery time ($\tau(fr)$, measured in days) at the ASCE 7 Design Earthquake hazard level (S_{DS} = 1.0 g and S_{D1} = 0.6g) for each structural system and story height.

System	Stories	Design	Repair Cost (Mil USD)		$\tau(fr)$	
			μ	σ	μ	σ
RCMF	4	baseline	$6.28 M	$6.84 M	282	208
RCMF	4	nonstructural	$5.99 M	$7.18 M	245	225
RCMF	4	structural	$2.72 M	$4.31 M	205	177
RCMF	4	full	$2.15 M	$3.9 M	154	176
RCMF	8	baseline	$4.94 M	$7.99 M	221	215
RCMF	8	nonstructural	$4.52 M	$7.85 M	183	223
RCMF	8	structural	$3.54 M	$6.36 M	191	186
RCMF	8	full	$3.03 M	$5.88 M	148	194
RCMF	12	baseline	$5.71 M	$8.64 M	273	224
RCMF	12	nonstructural	$5.34 M	$8.32 M	242	235
RCMF	12	structural	$2.74 M	$4.88 M	191	176
RCMF	12	full	$2.38 M	$4.46 M	155	184
RCMF	20	baseline	$8.4 M	$11.11 M	319	241
RCMF	20	nonstructural	$6.66 M	$9.53 M	277	250
RCMF	20	structural	$4.49 M	$7.1 M	240	193
RCMF	20	full	$3 M	$5.14 M	193	196
RCSW	4	baseline	$1.12 M	$2.49 M	142	157
RCSW	4	nonstructural	$0.68 M	$1.79 M	88	133
RCSW	4	structural	$1.06 M	$2.17 M	145	153
RCSW	4	full	$0.43 M	$1.44 M	94	138
RCSW	8	baseline	$3.56 M	$5.26 M	208	168
RCSW	8	nonstructural	$3.09 M	$4.74 M	166	162
RCSW	8	structural	$2.22 M	$3.27 M	208	161
RCSW	8	full	$0.86 M	$1.4 M	158	162
RCSW	12	baseline	$4.81 M	$5.99 M	252	170
RCSW	12	nonstructural	$4.64 M	$6.2 M	214	189
RCSW	12	structural	$0.91 M	$1.76 M	156	160
RCSW	12	full	$0.4 M	$0.87 M	110	152
RCSW	20	baseline	$6.14 M	$8.64 M	285	202
RCSW	20	nonstructural	$5.29 M	$8.18 M	236	196
RCSW	20	structural	$2.75 M	$4.38 M	230	166
RCSW	20	full	$1.38 M	$3.03 M	163	171
SMF	4	baseline	$10.21 M	$13.63 M	268	139
SMF	4	nonstructural	$9.03 M	$12.82 M	237	138
SMF	4	structural	$2.15 M	$2.74 M	213	149
SMF	4	full	$1.26 M	$2.14 M	142	142
SMF	8	baseline	$3.87 M	$5.26 M	208	143
SMF	8	nonstructural	$3.48 M	$4.81 M	170	146
SMF	8	structural	$4.7 M	$5.54 M	254	154
SMF	8	full	$3.46 M	$4.57 M	202	157
SMF	12	baseline	$8.22 M	$18.93 M	287	157
SMF	12	nonstructural	$8.34 M	$20.53 M	254	164
SMF	12	structural	$3.4 M	$5.42 M	255	163

System	Stories	Design	Repair Cost (Mil USD)		$\boldsymbol{\tau(fr)}$	
			μ	σ	μ	σ
SMF	12	full	$2.32 M	$4.77 M	199	170
SMF	20	baseline	$7.8 M	$7.75 M	314	151
SMF	20	nonstructural	$6.21 M	$6.81 M	278	169
SMF	20	structural	$4.79 M	$6.68 M	258	169
SMF	20	full	$3.4 M	$5.34 M	222	180
BRBF	4	baseline	$3.05 M	$5.22 M	260	146
BRBF	4	nonstructural	$2.3 M	$5.47 M	221	152
BRBF	4	structural	$2.42 M	$2.75 M	261	143
BRBF	4	full	$1.34 M	$2.05 M	219	148
BRBF	12	baseline	$3.4 M	$5.61 M	268	156
BRBF	12	nonstructural	$2.95 M	$6.65 M	228	167
BRBF	12	structural	$3.02 M	$3.48 M	263	157
BRBF	12	full	$1.74 M	$1.97 M	220	166
BRBF	18	baseline	$4.92 M	$5.4 M	299	156
BRBF	18	nonstructural	$2.39 M	$2.76 M	252	173
BRBF	18	structural	$5.12 M	$5.61 M	300	157
BRBF	18	full	$2.44 M	$2.84 M	251	175

5. Conclusion

This report documents the development of a portfolio of building archetype case studies for evaluating the seismic and, ultimately, economic performance of recovery-based design relative to baseline code-confirming design (i.e., ASCE 7-16) for new construction. We explore four lateral force resisting systems archetypes and several heights (number of stories) in applying structural and nonstructural recovery-based design interventions. Since the initial development of the archetype designs, updated versions of the ASCE 7 and applicable material design standards have been released, and design demand loads in southern California have increased by approximately 10 % to 20 %. Nevertheless, the overall findings of this study remain valid, as most conclusions are based on comparative analysis rather than absolute values.

We use the case studies to estimate construction costs using two approaches: one based on commercially available construction cost data and the other based on common industry best practices. The goal is not to provide definitive cost estimates but rather to benchmark the former approach against the latter. The results suggest that, conditional on similar project scopes, cost estimates based on commercially available data are reasonable approximations to cost estimates obtained through industry practices, but only in a limited scope of the estimation process. They may provide a reasonable first approximation depending on project scope and goals, but a professional cost engineer enhances the estimate through their ability to make informed assumptions about details of the project which may not yet be defined, and access to cost data sources beyond commercial cost databases.

Using the cost estimates derived from industry practice, we then simulate post-earthquake functional recovery performance for each of our archetype designs. The cost estimates are used to ground the estimates of repair costs. The results suggest substantial direct savings from recovery-based design in terms of repair costs. Moreover, annualized functional recovery performance suggests order-of-magnitude improvements in many cases.

5.1. Next Steps

Evaluating the tradeoff in enhanced post-earthquake performance and the associated cost increases is beyond the scope of this report. The archetype designs, cost estimates, and performance results are combined into a complete economic evaluation in Fung et al. (2025) [1].

5.2. Uncertainty

Uncertainty is an inherent aspect of engineering projects, particularly in civil engineering and cost engineering, where complex systems, environmental factors, and human variables interact. The cost estimates based on commercially available data use commercial cost data and FEMA P-58, as well as cost premiums suggested by manufacturers and engineers. This cost data is one potential source of uncertainty in this report. FEMA P-58 repair cost data is not updated every year to capture the changed markets for materials and labor. However, a small change in prices

can lead to inaccurate cost estimates, which in turn lead to cost overruns. In addition, commercial cost data providers do not generally publish the distribution statistics of their cost data, and there is not great transparency into their data collection process for regular users.

Sources of uncertainty in our cost estimation process include uncertainty for some project definitions and specific site conditions, including soil conditions and others. Project definition and cost estimation uncertainties were reduced by making documented assumptions where possible and critically analyzing cost data using multiple sources. Site condition uncertainty was also reduced by making documented assumptions for unknowns, using engineering judgment and using typical site conditions expected in the region of interest. Expert judgment may be subject to bias due to insufficient information, overconfidence, and other reasons,[5] which is an important source of uncertainty in cost estimation. To the extent that this uncertainty is still present in our cost estimates, an ASTM classification system of estimate processes is utilized to understand expected levels of uncertainty given our estimate class.

Cost estimates based on industry costs are subject to uncertainty as well. However, significant effort was expended into refining and validating the costs included herein, including seeking industry guidance and questioning and modifying assumptions as necessary. Through our method, and with the same commercial input data, the estimation process is repeatable and deterministic based on design details. To the extent that assumptions are made, these assumptions are documented to further support the repeatability and applicability of this estimation process to relevant projects. However, some variability in different professional and commercial cost estimates may be anticipated if different data sources are utilized. A variance in the cost estimation could be expected as a function of the variance of the input cost data. Thus, the methodology provides a defensible foundation for evaluating cost deltas and for using such cost deltas as inputs into a benefit-cost analysis (BCA). This is documented in Fung et al. (2025) [1].

Finally, the engineering models that are used to help determine expected damage, repair costs, and even functional recovery time are subject to typical material engineering and structural engineering uncertainty sources, many of which are explicitly considered in the performance assessment. Sources of uncertainty that are explicitly considered include variation in material properties, ground motion uncertainty, structural modeling uncertainty, construction quality, component-level damage and consequence functions, and uncertainties in impeding times delaying the start of repairs. Sources of uncertainty that are not considered, but may impact performance, include variations in member geometry and detailing based on design approach, variations in building geometry and configuration, uncertainty in building occupant tolerance for damage and adaptations to loss of function, and repair priority and sequencing uncertainty. The FEMA P-58 component-level fragility functions used in this study are empirical and may not generalize well to all structures or configurations. Additionally, cost and performance results

[5] Evaluating Uncertainty in Cost Estimates: A Practical Guide to Eliciting Expert Judgments. https://www.rand.org/pubs/research_briefs/RB216.htm

may not generalize well to other structures that are substantially different than those considered here.

5.3. Future Work

As stated earlier, this report provides the foundation for the full economic evaluation documented in Fung et al. (2025) [1]. In addition, we are extending the methods to estimate costs for recovery-based design options that are being actively considered for inclusion in guidance, codes, and standards [47]. Finally, while the first step in implementing recovery-based design is new construction, it is important to address the existing building stock. We are currently working on extending the methods to examine the benefits and costs of existing buildings retrofitted for functional recovery.

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